REP-114 · Standing orders: highest bid $2 higher and stake $2 lower, measured

Game	Fruit Market 1.20.0 live, rules version 18
Report date	2026-11-12 (Asia/Kuala_Lumpur)
Type	Analysis for a decision
Request	Owner, 2026-11-12: "for standing order add +2 to max bid and reduce stake by -2 for all standing order cards, how would this affect balance and fix mango orders"
Approval	Nothing is changed. Measured only; the owner decides (section 5).

1. The proposal

Order	Highest bid	Stake
Blueberry	$9 → $11	$6 → $4
Watermelon	$12 → $14	$8 → $6
Apple	$11 → $13	$7 → $5
Mango	$18 → $20	$12 → $10

Measured with the rules 18 balance scripts on the same seeds as REP-113 (patch `orderPlus2`): 1,500 games each at 3, 4 and 6 stalls with Normal computers, 1,000 at 4 stalls with Hard computers, and the paired test of bidding styles at 3 and 4 stalls (600 games a row).

2. Mango orders: a little better, not fixed

	3 stalls	4 stalls	6 stalls	4 stalls, Hard
Mango orders nobody bids for, now → proposed	44% → 38%	56% → 52%	49% → 45%	55% → 50%
Mango orders delivered in full	65% → 67%	51% → 56%	39% → 39%	66% → 66%
Stake lost for each mango order held	$3.71 → $2.95	$5.01 → $3.80	$6.29 → $5.11	$3.75 → $3.10

The cause is untouched: a mango order still needs the stake and the Green mangoes on the same day, and Green mangoes are scarce. $2 off the stake moves about one order in twenty.

3. The other orders: the higher top bid does almost nothing

Rivals' bids set the price, not the top bid. At 4 stalls the winning watermelon bid is $7.82 now and $7.79 with the proposal; blueberry $6.29 and $6.25; apple $7.44 and $7.41. Orders with no bidder fall a little (all orders at 4 stalls: 17% → 16%). Stake lost for each order held falls (all orders at 4 stalls: $1.79 → $1.27).

4. What bidding is worth, and balance

Seat made to bid one way, against bidding as the computer does	3 stalls now → proposed	4 stalls now → proposed
Always passes	−1.57 → −2.34	−0.88 → −1.66
Always bids the highest bid	−0.93 → −0.17	−0.80 → −0.64
Always bids 60% of the highest bid	−6.58 → −3.47	−3.84 → −2.03
Always bids half	−10.20 → −4.95	−7.95 → −1.45
Always bids $1	−28.92 → −29.04	−31.78 → −30.34

- Careful bidding becomes worth a little more ($1.70 to $2.30 a game over never bidding, from $0.90 to $1.60).
- Low bids are punished much less: a stall that always bids half the highest bid loses $1.45 a game at 4 stalls instead of $7.95. The stake stops reckless bids less well.
- **Shopkeepers:** no clear change. The largest distance from a fair win rate moves 3.6 → 5.1 points at 3 stalls, 4.6 → 2.9 at 4, 4.6 → 5.0 at 6 and 3.9 → 5.7 with Hard computers, each within about two standard errors (1.1 to 2.0 points a shopkeeper). Nest Egg and Bargain Hunter stay the strongest; Haggler and Shade Cloth the weakest.

5. Recommendation and options

The proposal makes orders a little more forgiving and a little more worth taking, but it does not fix mango orders, and it weakens the stake as a guard against wild bids. If the aim is mango orders, a different lever is needed. Options for the owner:

1. **Keep today's numbers** and add the sums on the bid sheet (REP-113 option 2, no rule change).
2. **Take the proposal as it stands** (rules version 19): small gain, mango still about half unbid.
3. **Fix mango orders directly** (rules version 19), each to be measured before choosing: a. a mango order is due in 3 days, so the mangoes can be bought the day after winning; b. a mango order wants one piece fewer (2, 3 and 4 by table size); c. replace the two mango order cards with one watermelon and one apple card.
4. **Combine**: the stake −$2 only (leave the highest bid), with one of 3a to 3c for mango.

6. How the numbers were made

`docs/studio/review-17/balance/scripts18: sim.ts --patch orderPlus2, analyze.ts --sections 1,7, focal.ts --test orders --patch orderPlus2`; baselines from REP-113, same seeds. Computer players only. Evidence in the session's scratchpad folder `so/`.